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International Cybersecurity Of Digital Forensic Academy (ICDFA)

PROGRAMME: Advance Critical Infrastructure Security (ACIS)

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Course: ACICP201 - Network Security Engineering

**Assignment: Week 2 Lab 1: Firewall Log Backup, Event
Interpretation and Network Operations Report**
**Firewall logging, log backup, operational visibility, and network
evidence handling**

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Task 1 – Firewall Log Backup Plan

1.1 Introduction

Firewall logs are an essential component of security monitoring in critical infrastructure environments. They provide evidence of network activity, policy enforcement, security incidents, administrator actions, and system health. A structured backup and retention strategy ensures that firewall logs remain confidential, accurate, available for investigations, and compliant with legal and regulatory requirements. This backup plan follows guidance from NIST SP 800-41 Rev.1, NIST SP 800-82 Rev.3, and NIST SP 800-92.

1.2 Firewall Log Sources

Log Source	Description	Purpose
Internet Edge Firewall	Monitors inbound and outbound Internet traffic	Detect attacks, malware, scanning, unauthorized access
Internal Segmentation Firewall	Controls communication between corporate LAN and OT network	Prevent unauthorized lateral movement
VPN Gateway	Records remote user authentication and VPN sessions	Monitor remote access activities
DMZ Firewall	Protects public-facing servers	Monitor web attacks and DMZ communications
Engineering Jump-Host Firewall	Controls administrator access to OT systems	Audit privileged access
SIEM / Log Collector	Central repository for all security logs	Correlation, reporting and long-term storage

1.3 Firewall Log Types

The following log categories shall be collected.

Log Type	Description
Traffic Logs	Allowed and denied network connections
Threat Logs	IPS alerts, malware detection, exploit signatures
Authentication Logs	User login attempts, MFA events
Configuration Change Logs	Firewall rule modifications and policy updates
Administrator Activity Logs	Administrator login/logout and configuration changes
System Health Logs	CPU usage, memory, interface status, hardware alerts
VPN Logs	VPN login, logout, session duration and failures

1.4 Log Transfer Method

Firewall logs shall be transmitted securely from each firewall to the centralized SIEM using encrypted communication.

Backup Workflow

Firewalls

Encrypted Syslog (TLS)

SIEM / Log Collector

Daily Incremental Backup

Weekly Full Backup

Encrypted Backup Repository

Cold Archive Storage

Security Controls:

- TLS-encrypted Syslog forwarding.
- AES-256 encrypted backup files.
- Role-Based Access Control (RBAC).
- Multi-Factor Authentication for administrators.
- SHA-256 integrity verification.
- Write-once archive storage where applicable.

1.5 Log Retention Policy

Storage Tier	Retention Period	Purpose
Hot Storage	30 days	Active monitoring and incident response
Warm Storage	60 additional days	Operational investigations
Cold Archive	365 days	Compliance and historical analysis
Legal Hold	Until investigation concludes	Preserve evidence for legal proceedings

1.6 Backup Schedule

Backup Activity	Frequency
Incremental Backup	Daily
Full Backup	Weekly
Archive Review	Monthly
Restore Test	Quarterly

1.7 Evidence Protection Measures

- Firewall logs are treated as digital evidence.
- The following controls are implemented:
- SHA-256 hashing after every backup.
- Network Time Protocol (NTP) synchronization for consistent timestamps.
- Chain-of-custody documentation for all exported logs.
- Read-only permissions on archived logs.
- Multi-factor authentication for backup administrators.
- Continuous integrity verification to detect tampering.
- Audit logs recording every access to archived firewall logs.

1.8 Roles and Responsibilities

Role	Responsibility
Network Security Engineer	Configure firewall logging
SOC Analyst	Monitor firewall events
Backup Administrator	Perform scheduled backups
SIEM Administrator	Maintain centralized logging
Incident Response Team	Investigate security events
Compliance Officer	Verify retention and legal requirements

Task 2 – Log Storage and Retention Calculations

Introduction

Proper firewall log storage planning ensures that security events remain available for incident response, forensic investigations, compliance audits, and regulatory requirements. The following calculations estimate storage requirements using the fictional firewall log volumes provided in the assignment.

A. Calculate the Total Daily Log Volume

Question	Calculation Requirement
A Total Daily Volume=2.8+1.6+0.7+0.9 Working =6.0 GB/day Answer : Total Daily Firewall Log Volume = 6.0 GB/day	Edge firewall generates 2.8 GB/day. Internal firewall generates 1.6 GB/day. VPN gateway generates 0.7 GB/day. DMZ firewall generates 0.9 GB/day. Calculate total daily log volume.
B Compressed Size=6.0×40% Working=6.0×0.40 =2.4 GB/day Answer: Compressed Daily Storage = 2.4 GB/day	If logs are compressed at a 60% reduction rate, calculate compressed daily storage requirement.
C 90×2.4 Working =216 GB Answer: 90-Day Compressed Storage = 216 GB	Calculate compressed storage required for 90 days of hot/warm retention.
D 365×2.4 Working =876 GB Answer:365-Day Archive Storage = 876 GB	Calculate compressed storage required for 365 days of archive retention.
E 876×18	If archive storage costs ₦18 per GB per month, estimate the monthly storage cost for 365-day archive volume.

Working =15,768 Answer:Estimated Monthly Archive Storage Cost = ₦15,768	
F Growth Rate:12% every quarter Two quarters:12%+12%=24% Original Daily Volume 6.0 GB Formula 6.0×1.24 Working =7.44 GB/day Answer:Estimated Daily Log Volume After Two Quarters = 7.44 GB/day	If log growth increases by 12% per quarter, estimate the daily uncompressed log volume after two quarters.
G Restore Speed 35 minutes per 100 GB Assumption 1 TB = 1000 GB Formula $1000 \div 100 = 10$ 10×35 Working =350 minutes Convert to hours $350 \div 60 = 5.83$ hours =0.83 hour $0.83 \times 60 \approx 50$ minutes Answer Estimated Restore Time = 350 minutes (approximately 5 hours 50 minutes) Assumption: The calculation uses the decimal storage convention (1 TB = 1000 GB) and assumes a constant restore speed with no interruptions or network bottlenecks.	If a restore test takes 35 minutes per 100 GB, estimate the time needed to restore 1 TB of archived logs. State any assumptions.

Task 3 – Event Interpretation

Introduction

Firewall event interpretation enables security teams to distinguish between legitimate network activity and events that require investigation or immediate response. In a critical infrastructure environment, such as an Industrial Control System (ICS) or SCADA network, unauthorized communications between network zones may pose operational and safety risks. The following table interprets the fictional firewall events provided in the assignment using the prescribed severity levels.

Severity Guide

Critical: Likely active compromise or dangerous OT policy violation.

High: Urgent security concern requiring immediate investigation.

Medium: Suspicious activity or policy exception.

Low: Expected operational event or informational log.

Event Interpretation Table

ID	Event Type	Likely Meaning	Severity	Required Response	Escalate?
E01	Unauthorized Modbus Communication	A corporate workstation attempted to communicate with an OT device using the Modbus protocol (TCP/502). The firewall correctly blocked the connection because the source device was not approved. This indicates proper network	Critical	Verify whether the attempt was authorized. Investigate the source workstation for possible malware, misconfiguration, or unauthorized user activity. Confirm firewall policy remains effective.	Yes – Immediately

ID	Event Type	Likely Meaning	Severity	Required Response	Escalate?
E02	Approved Remote Administrative Access	segmentation. A remote engineer successfully authenticated using Multi-Factor Authentication (MFA) during an approved maintenance window and accessed the engineering jump server through SSH. This is expected administrative activity.	Low	Record the successful login, verify change request documentation, and continue routine monitoring of the session.	No
E03	Web Attack Detected Blocked	An Internet host attempted to exploit a web application in the DMZ. The firewall and detected an exploit signature, allowed initial inspection, then reset the session to stop the attack.	High	Review IPS logs, inspect the web server for signs of compromise, update threat signatures if required, and monitor for repeated attempts from the same source.	Yes
E04	Administrator Login Outside Maintenance Window	A firewall administrator successfully logged in outside the organization's approved maintenance schedule. Although authentication succeeded, the timing represents a policy exception that requires verification.	Medium	Contact the administrator to verify the activity, review change management records, and examine administrator actions performed during the session.	Yes
E05	Blocked Internet Access from OT Network	An OT workstation attempted to establish an outbound HTTPS connection to an Internet address.	Critical	Immediately investigate the OT workstation for malware, unauthorized software, or	Yes Immediately

ID	Event Type	Likely Meaning	Severity	Required Response	Escalate?
E06	Normal Syslog Forwarding	Firewall policy correctly blocked the connection to protect the industrial control environment.	Low	configuration errors. Verify the allowlist and inspect endpoint logs.	No
		The DMZ server successfully forwarded firewall logs to the SIEM using Syslog (UDP/514). This confirms normal log collection and centralized monitoring.		No corrective action required. Continue monitoring to ensure consistent log delivery.	
		A corporate workstation uploaded approximately 950 MB of data to an Internet destination within eight minutes. This could represent legitimate business activity or possible data exfiltration.		Identify the user and destination, verify business justification, inspect Data Loss Prevention (DLP) alerts, and review endpoint activity for suspicious behaviour.	
E07	Large Outbound Data Transfer		High		Yes
E08	Unauthorized RDP Connection Attempt	A jump-host attempted Remote Desktop access to an engineering workstation that was not included in the firewall allowlist. The firewall correctly denied the connection.	High	Verify whether the connection was part of an approved maintenance task, review firewall rules, and investigate any unauthorized access attempts.	Yes

Event Summary

Severity Level	Number of Events
Critical	2
High	3
Medium	1
Low	2

Priority Analysis

Critical Events

The most significant security concerns are E01 and E05 because they involve attempts to access or communicate with the OT/SCADA environment outside approved policies. Such events could indicate malware propagation, unauthorized lateral movement, or attempts to compromise industrial control systems. Immediate investigation is necessary to maintain operational safety and prevent disruption.

High-Severity Events

Events E03, E07, and E08 require prompt investigation. The detected web exploit (E03) suggests an external attack against a public-facing service, the large outbound data transfer (E07) may indicate possible data exfiltration, and the unauthorized RDP attempt (E08) represents a violation of access control policies within the engineering environment.

Medium-Severity Event

Event E04 reflects a successful administrator login outside the approved maintenance window. Although not necessarily malicious, it represents a deviation from established operational procedures and should be verified against approved change requests.

Low-Severity Events

Events E02 and E06 represent expected operational behaviour. The successful VPN login during an approved maintenance window demonstrates proper access controls, while normal Syslog forwarding confirms that centralized log collection is functioning correctly.

Task 4 – Evidence Handling and Chain of Custody

4.1 Introduction

Firewall logs are valuable digital evidence that support incident response, forensic investigations, compliance audits, and legal proceedings. To preserve their evidential value, logs must be collected, stored, transferred, and retained in a manner that maintains their confidentiality, integrity, availability, authenticity, and accountability. A documented chain-of-custody process ensures that firewall log evidence remains admissible and trustworthy throughout its lifecycle.

4.2 Firewall Log Evidence Handling Process

The following process describes how firewall logs should be securely managed within a critical infrastructure environment.

Step 1: Log Collection and Export

Firewall logs are automatically generated by the following systems:

Internet Edge Firewall

Internal Segmentation Firewall

VPN Gateway

DMZ Firewall

Engineering Jump-Host Firewall

SIEM/Log Collector

Logs are securely forwarded to the centralized SIEM using TLS-encrypted Syslog.

Daily exports are created in a read-only format for backup and forensic purposes.

Purpose

Centralize security monitoring.

Prevent loss of log data.

Support forensic investigations.

Meet regulatory compliance requirements.

Step 2: Protection of Original Logs

Original firewall logs must never be modified after collection.

Protection measures include:

Store original logs in a read-only repository.

Restrict modification permissions.

Encrypt backups using AES-256.

Apply Role-Based Access Control (RBAC).

Enable Multi-Factor Authentication (MFA) for administrators.

Record all access in audit logs.

These controls ensure that firewall evidence remains authentic and untampered.

Step 3: Hash Generation and Integrity Verification

Immediately after exporting firewall logs, a cryptographic hash value should be generated.

Recommended algorithm:

SHA-256

Example:

Firewall_Log_2026-08-10.zip

SHA-256:

4F98A7C13B5E82D2D7F7E9A7C4D8B81A7A1C4A8B6E9F5D3A4C2D9F6E7A1
B2C3

The generated hash value should be recorded in:

Evidence Register

Chain-of-Custody Form

Incident Ticket

SIEM Case Record

Whenever logs are transferred or restored, the hash must be recalculated and compared with the original value.

Matching hashes confirm that the evidence has not been altered.

Step 4: Evidence Repository Access Control

Access to firewall evidence should follow the Principle of Least Privilege.

Role	Access Level
Security Operations Centre (SOC) Analyst	Read Only
Incident Response Team	Read Only / Export
Network Security Engineer	Read and Manage Backups
SIEM Administrator	Administrative Access
Compliance Officer	Audit Access
Legal Team	Read Only (Approved Cases)

All evidence access should be logged automatically.

Step 5: Time Synchronization Verification

Accurate timestamps are essential during incident investigations.

The organization should synchronize all systems using an approved Network Time Protocol (NTP) server.

Systems requiring synchronization include:

Firewalls

SIEM Server

VPN Gateway

Domain Controllers

Engineering Jump Hosts

OT Servers

Time synchronization should be verified daily.

Benefits include:

Accurate event correlation.

Reliable forensic timelines.

Consistent incident reporting.

Improved audit compliance.

Step 6: Evidence Transfer Process

When firewall evidence is required for investigation or legal purposes, the following process should be followed:

Export required firewall logs.

Generate SHA-256 hash.

Record evidence details.

Encrypt evidence archive.

Transfer through an approved secure channel.

Verify hash after receipt.

Obtain recipient acknowledgement.

Update the Chain-of-Custody Register.

Recipients may include:

Incident Response Team

Digital Forensics Team

Legal Department

Compliance Office

Executive Management

Step 7: Secure Evidence Disposal

When the retention period expires and there is no legal hold, firewall log evidence should be securely destroyed.

Approved methods include:

Cryptographic erasure.

Secure overwrite of storage media.

Secure deletion following organizational policy.

Destruction approval documented by the Compliance Officer.

The disposal record should include:

Evidence ID

Date destroyed

Disposal method

Authorizing officer

Witness (if required)

4.3 Chain-of-Custody Record

Evidence ID	Description	Date & Time Collected	Collected By	Hash (SHA-256)	Stored Location	Transferred To	Date & Time	Signature
FW-2026-001	Edge Firewall Logs	10-Aug-2026 10:30	SOC Analyst	SHA-256 Recorded	Secure Evidence Repository	Incident Response Team	10-Aug-2026 11:15	Digital Signature
FW-2026-002	VPN Authentication Logs	10-Aug-2026 10:45	SOC Analyst	SHA-256 Recorded	Secure Evidence Repository	Compliance Officer	10-Aug-2026 12:00	Digital Signature

4.4 Evidence Lifecycle

Firewall Generates Logs

Secure Collection (TLS Syslog)

Central SIEM / Log Collector

Generate SHA-256 Hash

Encrypt Backup

Secure Evidence Repository

Access Control & Audit Logging

Incident Response / Legal Review

Retention Period Expires

Secure Disposal & Documentation

4.5 Best Practices for Firewall Evidence Handling

The following practices should be adopted:

- Synchronize all systems using trusted NTP servers.
- Generate SHA-256 hashes for every exported log.
- Encrypt backups using AES-256.
- Restrict evidence access using RBAC and MFA.
- Maintain a complete chain-of-custody record.
- Conduct quarterly restore and integrity verification tests.
- Store backups in geographically separate locations.
- Retain logs according to organizational and regulatory requirements.
- Monitor all evidence access through audit logs.
- Securely destroy expired evidence following approved procedures.

Task 5: Mini Network-Operations Report

Executive Summary

This report presents the results of a firewall log backup and security monitoring assessment for a fictional critical infrastructure environment. The assessment focuses on firewall log management, storage, event interpretation, evidence handling, and operational recommendations. Firewall logs were assumed to be collected from the Internet edge, internal segmentation firewall, VPN gateway, DMZ firewall, engineering jump zone, and OT network segment before being securely forwarded to a centralized Security Information and Event Management (SIEM) platform.

Eight fictional firewall events were analyzed to distinguish between normal operational activities and suspicious or policy-violating behaviour. The assessment identified two Critical, three High, one Medium, and two Low severity events. The most significant risks involved unauthorized attempts to access the OT/SCADA network, detected web exploitation attempts, and unusually large outbound data transfers from the corporate network.

The report concludes that while firewall segmentation and centralized logging provide strong defensive controls, improvements are required in administrative monitoring, data loss detection, and continuous firewall policy review. Recommended actions include immediate investigation of high-risk events, enhanced monitoring, periodic firewall rule reviews, and regular disaster recovery testing of firewall log backups.

1. Network Zones Reviewed

The assessment covered the following security zones:

Network Zone	Purpose	Primary Security Function
Internet Edge	External connectivity	Protects the organization from Internet threats and performs NAT, IPS, and anti-spoofing inspection.
DMZ	Public-facing services	Hosts web servers and external applications while isolating them from internal networks.
Corporate LAN	Business users	Provides access to enterprise resources while enforcing user-based security policies.
VPN Zone	Remote access	Provides secure encrypted access for authorized remote users using MFA.
Engineering	Administrative	Provides controlled administrative access to OT

Network Zone	Purpose	Primary Security Function
Jump Zone	access	systems through approved jump hosts.
OT/SCADA Network	Industrial control systems	Supports critical operational technology and industrial process control using strict allowlist policies.

2. Summary of Log Backup Status

The organization's firewall log backup strategy provides centralized collection, secure storage, and long-term retention of security events.

The review found that:

- Firewall logs are forwarded securely to the SIEM using encrypted Syslog.
- Daily incremental backups are performed.
- Weekly full backup exports are generated.
- Monthly archive verification is conducted.
- Quarterly restore testing verifies backup integrity.
- SHA-256 hashing protects evidence integrity.
- Access to archived logs is restricted using Role-Based Access Control (RBAC).
- Backup data is encrypted using AES-256.
- Network devices synchronize time through centralized NTP servers.
- Overall, the backup process satisfies the confidentiality, integrity, and availability (CIA) requirements expected in a critical infrastructure environment.

3. Top Three Security Events

Event 1 – Unauthorized Corporate Access to OT Network (E01)

A workstation in the Corporate LAN attempted to communicate with an OT/SCADA device using the Modbus protocol (TCP/502). The firewall denied the connection because the workstation was not part of the approved allowlist.

Severity: Critical

Impact

This event demonstrates that network segmentation is functioning correctly. However, unauthorized attempts to access industrial control devices could indicate malware infection, insider misuse, or configuration errors.

Recommended Action

- Investigate the originating workstation.
- Review endpoint security logs.
- Confirm firewall policies remain unchanged.
- Scan the workstation for malware.

Event 2 – Web Exploit Attempt Against DMZ Server (E03)

A public Internet host attempted to exploit a web application hosted within the DMZ. The firewall detected the exploit signature and terminated the session before the attack could continue.

Severity: High

Impact

Although the firewall successfully blocked the attack, repeated exploit attempts may indicate targeted reconnaissance or automated vulnerability scanning.

Recommended Action

- Review IPS alerts.
- Verify web server integrity.

- Apply software updates if required.
- Continue monitoring similar attack patterns.

Event 3 – Large Outbound Data Transfer (E07)

A workstation within the Corporate LAN uploaded approximately 950 MB of data to an Internet destination within eight minutes.

Severity: High

Impact

The activity could represent legitimate business operations or possible data exfiltration. Without additional investigation, the organization cannot determine whether sensitive information was transferred.

Recommended Action

- Identify the user responsible.
- Verify business justification.
- Review Data Loss Prevention (DLP) alerts.
- Inspect endpoint audit logs.

4. Operational Impact and Risk

The firewall successfully enforced segmentation policies by preventing unauthorized communications between network zones. This significantly reduces the risk of lateral movement into the OT environment.

However, several events demonstrate areas requiring additional monitoring:

- Unauthorized access attempts toward OT assets could threaten operational continuity.
- Administrative activity outside approved maintenance windows increases governance risk.
- Large outbound file transfers could result in confidential information leakage.
- External web attacks continue to target publicly accessible services.
- If these risks are not properly managed, they may lead to service disruption, unauthorized access, regulatory violations, or compromise of industrial control systems.

5. Recommended Actions (Next 7 Days)

The following actions should be completed immediately:

- Investigate all Critical and High severity events.
- Verify the integrity of firewall configurations.
- Review VPN authentication logs.
- Confirm all administrator activities are supported by approved change requests.
- Investigate the large outbound data transfer.
- Review firewall rules protecting the OT network.
- Validate SIEM log collection from all firewall devices.
- Perform malware scans on systems involved in suspicious events.

6. Recommended Actions (Next 30 Days)

Longer-term improvements should include:

- Review firewall policies and remove obsolete rules.
- Implement Data Loss Prevention (DLP) monitoring for outbound traffic.
- Conduct a firewall rule recertification exercise.
- Perform a disaster recovery restore test for archived firewall logs.
- Review user privileges using the Principle of Least Privilege.

- Update intrusion prevention signatures.
- Conduct cybersecurity awareness training for administrators.
- Test incident response procedures using simulated firewall events.
- Validate backup integrity through SHA-256 verification.
- Review log retention policies for compliance with organizational requirements.

7. Conclusion

This assessment demonstrates the importance of effective firewall monitoring, centralized logging, and secure evidence management within a critical infrastructure environment. The firewall successfully blocked unauthorized communications, detected malicious activity, and maintained network segmentation between corporate and operational technology networks.

Although the organization has implemented strong foundational controls, several security events require prompt investigation and continuous monitoring. Improving administrative oversight, reviewing firewall policies, strengthening data loss prevention, and routinely testing backup recovery processes will further enhance the organization's cybersecurity posture.

By maintaining secure log backups, preserving evidence integrity through cryptographic hashing, enforcing least-privilege access, and continuously monitoring firewall activity, the organization can improve operational resilience, support regulatory compliance, and ensure the continued protection of critical infrastructure services.